CONOR BROWN

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EDUCATION

Vanderbilt University — Ph.D., Electrical & Computer Engineering Nashville, TN | Aug 2025 – Expected 2030
Current - GPA 4.0 / 4.0 · Best Teaching Assistant ECE 2026

University of Tennessee, Knoxville — B.S., Biomedical Engineering Knoxville, TN | May 2023
GPA 4.0 / 4.0 · Top Graduate & Outstanding Senior Award

RESEARCH EXPERIENCE

Graduate Research Assistant — Vanderbilt University, Noble Lab Nashville, TN | Aug 2025 – Current

- Developed differentiable computational models of the full auditory pathway: an ERB-spaced gammatone / Meddis cochlear model, a clinical cochlear-implant signal path (ACE N-of-M coding and electrical spread of excitation), and a biophysical, patient-specific auditory-nerve model to quantify and improve how faithfully implants convey sound.
- Built an end-to-end differentiable encoder in PyTorch that maps audio directly to electrode selections and pulse currents, optimized per patient and exportable as a hardware-realizable pulse table.
- Designed convolutional decoders that reconstruct audio from simulated neural firing, and Transformer models predicting STFT phase from log-magnitude spectra, evaluated with MFCC-based metrics.

Research Assistant — UT SMART Lab, University of Tennessee Knoxville, TN | Feb 2023 – Jul 2025

- Built AI-driven digital-health tools for dementia care - a RAG chatbot (LangChain, OpenAI, Pinecone), assistive web applications, and social-robotics control systems, collaborating across interdisciplinary engineering and research teams.

Medical Assistant — Camai Community Health Center Naknek, AK | May 2022 – Aug 2022

- Performed phlebotomy, IV medication administration, wound debridement, radiography, suturing, and laboratory testing in primary and emergency care; served as the clinic's go-to resource for troubleshooting medical technology.

SELECTED RESEARCH PROJECTS

Personalized Cochlear Implant Encoding

A patient-specific, fully differentiable model of the auditory pathway that learns optimized electrical stimulation for an individual ear and exports it to a hardware-realizable pulse table. Built as a staged progression of models:

- **Healthy-hearing reference:** an ERB-spaced gammatone filter bank with Meddis inner-hair-cell transduction paired with a convolutional decoder that reconstructs the waveform, establishing an upper bound on the information a healthy cochlea preserves.
- **Cochlear-implant signal path:** an ACE (N-of-M) sound-coding model with electrical spread of excitation, mapped into the same neural representation so the degradation introduced by electrical stimulation is directly measurable.
- **Patient-specific auditory nerve:** a biophysical, per-pulse refractory nerve-fiber model driven by a digital twin of a patient's threshold map and clinical fitting, capturing how a pulse train actually translates into neural firing rather than assuming a linear response.
- **Learned encoder:** an end-to-end differentiable encoder (PyTorch) trained through the nerve model and decoder to map audio directly to electrode selections and pulse currents, respecting real clinical constraints (current limits, N-of-M selection, and stimulation rate) and exportable to a device-ready pulse table.

TECHNICAL SKILLS

Programming: Python, Rust, C, MATLAB, JavaScript

ML & Scientific Computing: PyTorch, deep learning, differentiable programming, digital signal processing, CUDA / Metal kernels, NumPy / SciPy